PSet 2: Health Insurance

```
library(tidyverse)

# Load data from .csv file
data_file <- if (file.exists("health-insurance.csv")) {
  "health-insurance.csv"
} else {
  list.files(pattern = "\\.csv$")[1]
}

# skip the 4 metadata lines at the top of the MDAT export
raw <- read_csv(data_file, skip = 4,
                col_types = cols(.default = col_character()))
names(raw) <- c("group", "age", "count")

# group rows look like "-> MAR -> HICOV -> TEN", age rows sit below them
cells <- raw |>
  mutate(count = parse_number(count)) |>
  fill(group) |>
  filter(!is.na(age), age != "Total") |>
  separate_wider_delim(group, "->", too_few = "align_start",
                       names = c(NA, "mar", "hicov", "ten")) |>
  filter(!is.na(ten)) |>
  mutate(
    across(c(mar, hicov, ten), str_squish),
    age = str_replace(age, "Between (\\d+) and (\\d+)", "\\1-\\2"),
    age_lo = parse_number(age),
    age = fct_reorder(age, age_lo),
    mar = str_remove(mar, " or under 15 years old") |>
      fct_relevel("Never married", "Married", "Separated", "Divorced"),
    coverage = if_else(str_starts(hicov, "No"), "Uninsured", "Insured"),
    ten = case_when(
      str_starts(ten, "N/A") ~ "Group quarters",
      str_detect(ten, "mortgage") ~ "Owned with mortgage",
      str_detect(ten, "Free") ~ "Owned free and clear",
      str_detect(ten, "without payment") ~ "No rent paid",
      .default = ten
    ) |>
      fct_relevel("Owned with mortgage", "Owned free and clear",
                  "Rented", "No rent paid")
  )

# weighted population and share uninsured for any grouping
rates <- function(data, ...) {
  data |>
    group_by(...) |>
    summarise(pop = sum(count),
              uninsured = sum(count[coverage == "Uninsured"]),
              .groups = "drop") |>
    mutate(rate = uninsured / pop)
}

coverage_cols <- c(Insured = "grey65", Uninsured = "#D55E00")
```

```
# colorblind-safe Okabe-Ito colors, skipping black, yellow and vermillion
tenure_cols <- unname(palette.colors()[c(2, 3, 4, 8, 6)])
```

```
cells |>
  ggplot(aes(age, count / 1e6, fill = coverage)) +
  geom_col() +
  scale_fill_manual(values = coverage_cols) +
  labs(title = "Population by age and coverage", x = "Age",
       y = "People (millions)", fill = "Coverage",
       caption = "Figure 1") +
  theme_minimal(12)
```

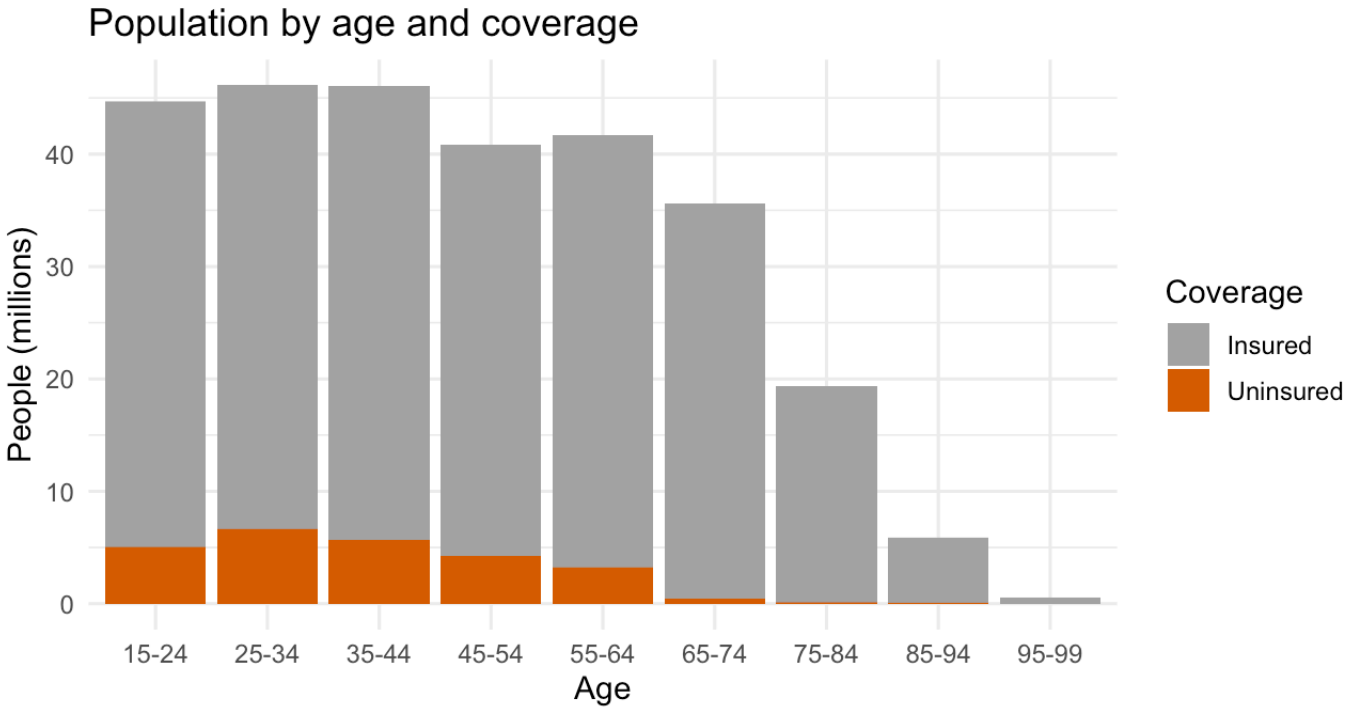


Figure 1

```
rates(cells, age) |>
  ggplot(aes(age, rate)) +
  geom_col(fill = "#D55E00") +
  scale_y_continuous(labels = scales::label_percent()) +
  labs(title = "Uninsured rate by age", x = "Age",
       y = "Uninsured", caption = "Figure 2") +
  theme_minimal(12)
```

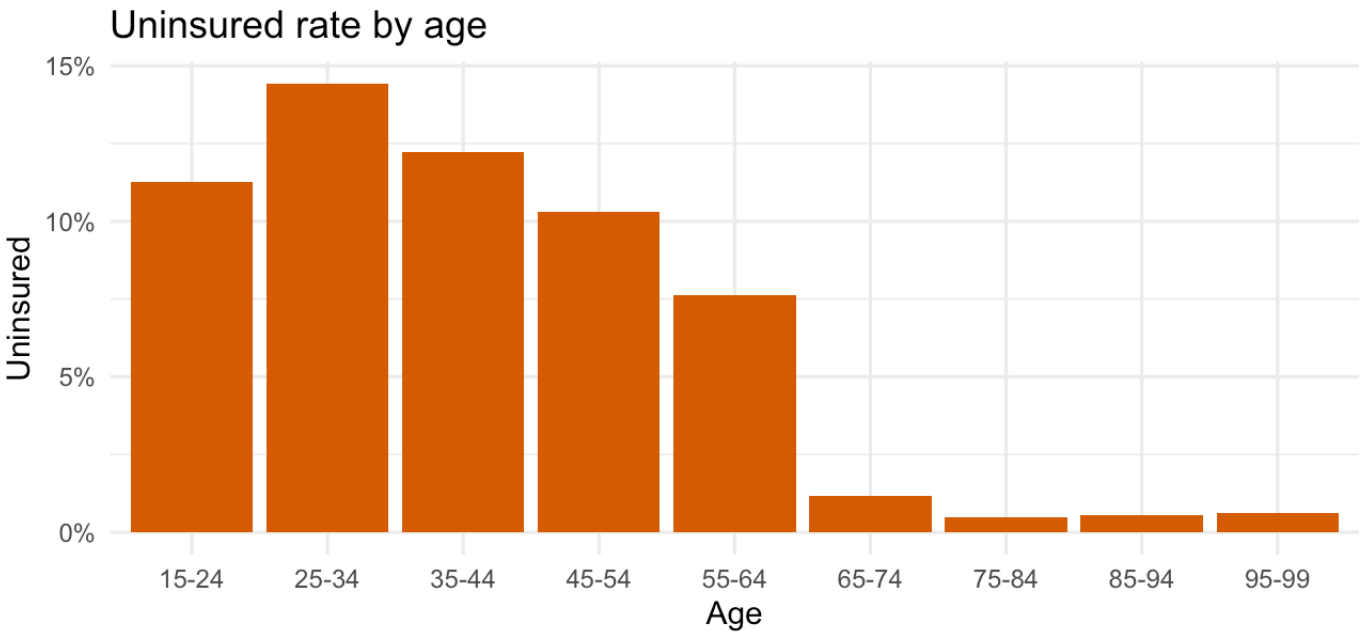


Figure 2

```
rates(cells, mar) |>
  ggplot(aes(rate, fct_reorder(mar, rate))) +
  geom_point(size = 3) +
  scale_x_continuous(labels = scales::label_percent(),
                     limits = c(0, NA)) +
  labs(title = "Uninsured rate by marital status (all ages)",
       x = "Uninsured", y = NULL, caption = "Figure 3") +
  theme_minimal(12)
```

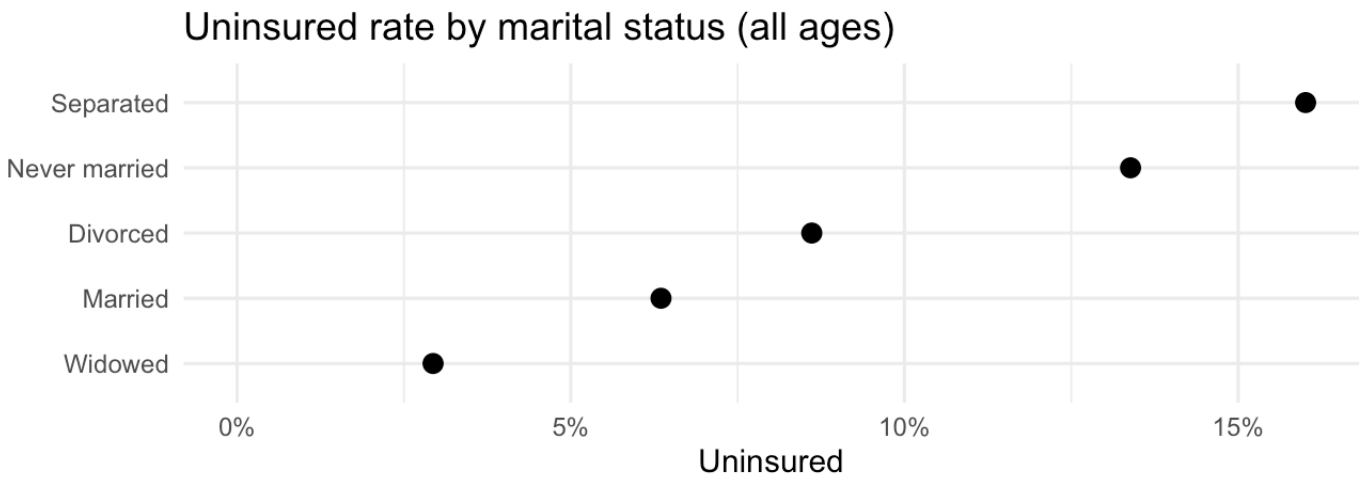


Figure 3

```
cells |>
  ggplot(aes(age, count, fill = mar)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_fill_brewer(palette = "Dark2") +
  labs(title = "Marital status by age", x = "Age",
       y = "Share of age group", fill = "Marital status",
```

```
      caption = "Figure 4") +
  theme_minimal(12)
```

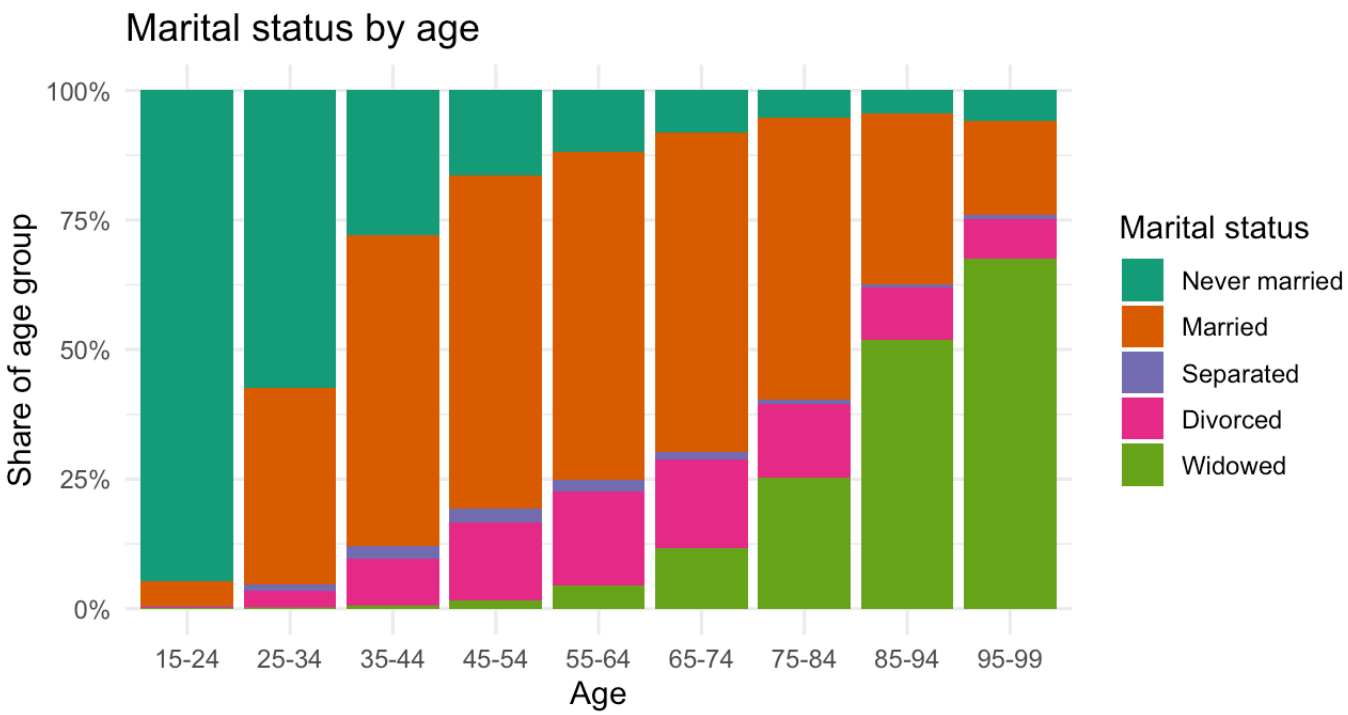


Figure 4

```
rates(cells, age, mar) |>
  ggplot(aes(age, rate, color = mar, group = mar)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_color_brewer(palette = "Dark2") +
  labs(title = "Uninsured rate by age and marital status",
       x = "Age", y = "Uninsured", color = "Marital status",
       caption = "Figure 5") +
  theme_minimal(12)
```

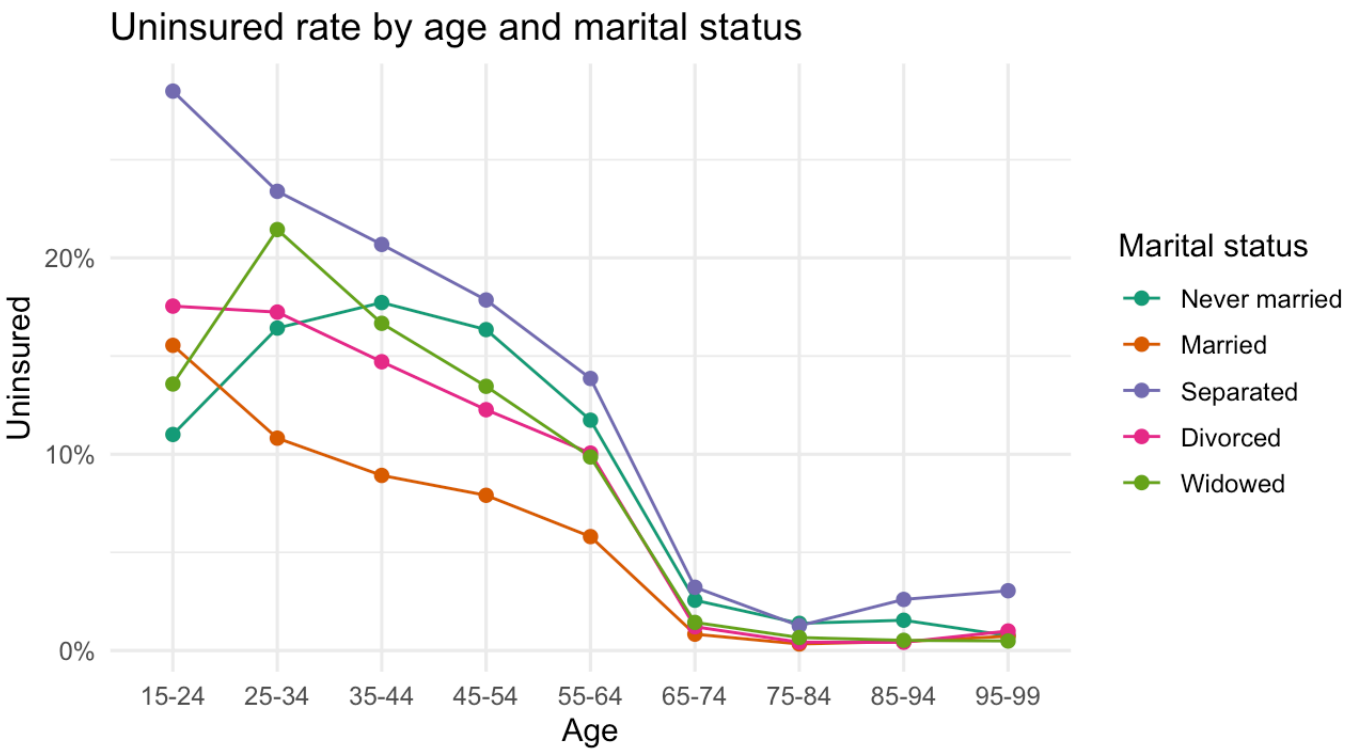


Figure 5

```
rates(cells, ten) |>
  ggplot(aes(rate, fct_reorder(ten, rate))) +
  geom_point(size = 3) +
  scale_x_continuous(labels = scales::label_percent(),
                     limits = c(0, NA)) +
  labs(title = "Uninsured rate by housing tenure (all ages)",
       x = "Uninsured", y = NULL, caption = "Figure 6") +
  theme_minimal(12)
```

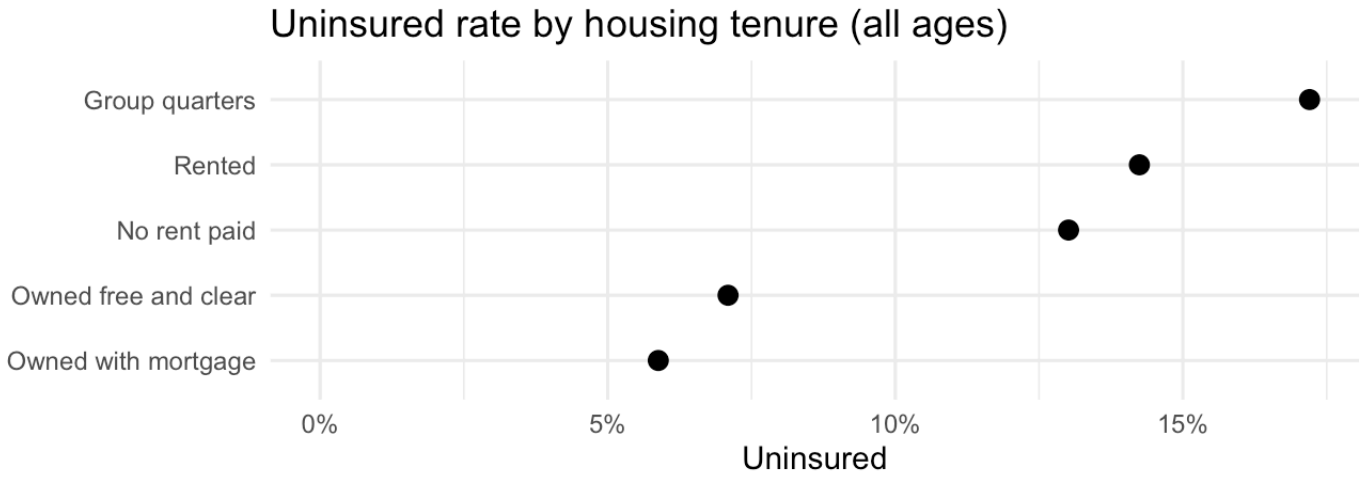


Figure 6

```
cells |>
  ggplot(aes(age, count, fill = ten)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_fill_manual(values = tenure_cols) +
```

```
  labs(title = "Housing tenure by age", x = "Age",
       y = "Share of age group", fill = "Tenure",
       caption = "Figure 7") +
  theme_minimal(12)
```

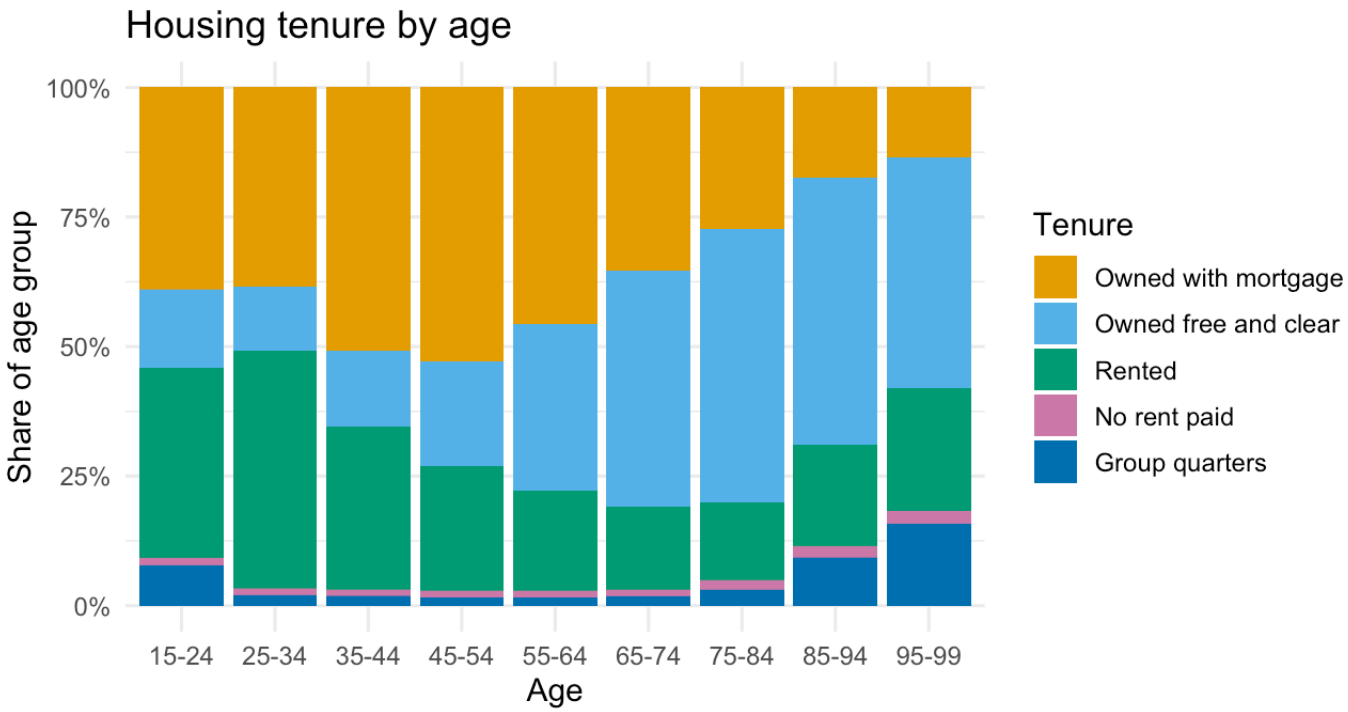


Figure 7

```
rates(cells, age, ten) |>
  ggplot(aes(age, rate, color = ten, group = ten)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_continuous(labels = scales::label_percent()) +
  scale_color_manual(values = tenure_cols) +
  labs(title = "Uninsured rate by age and housing tenure",
       x = "Age", y = "Uninsured", color = "Tenure",
       caption = "Figure 8") +
  theme_minimal(12)
```

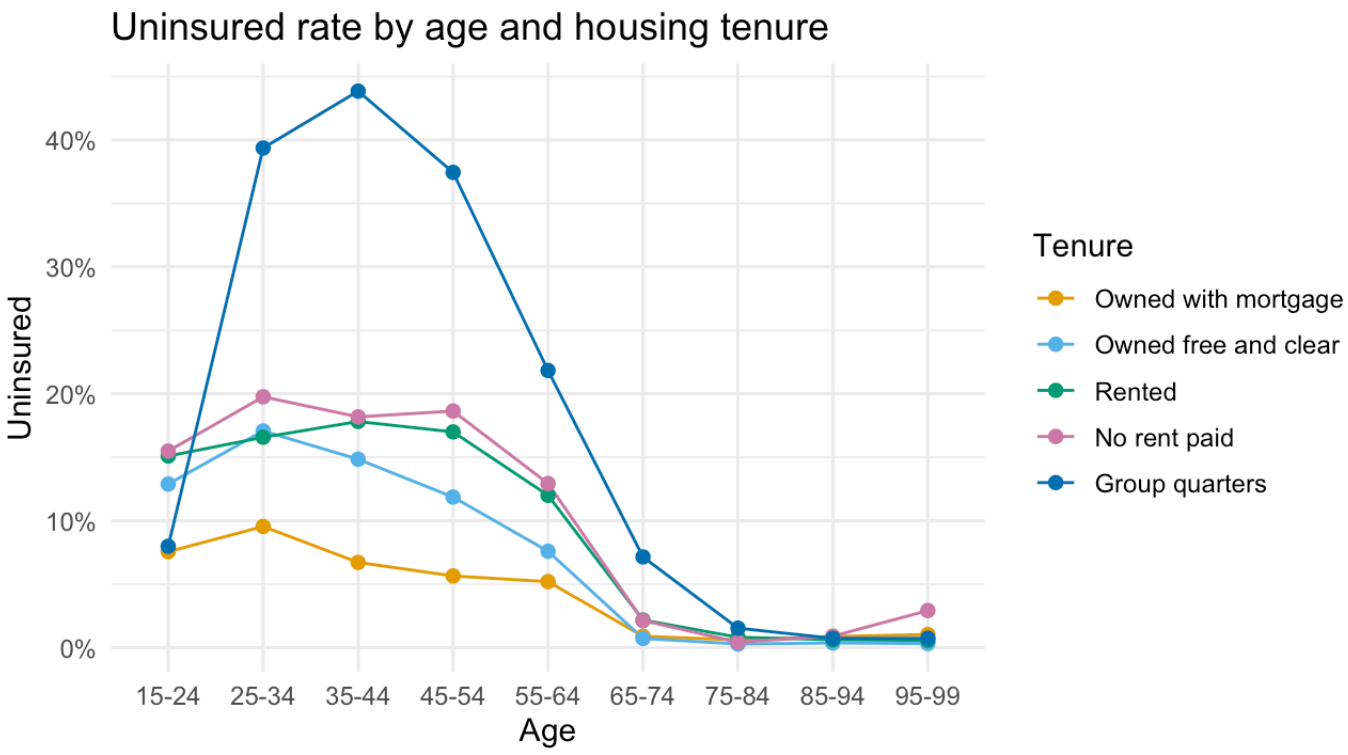


Figure 8

```
# ages 15-64 only, since almost everyone 65+ is insured
cells |>
  filter(age_lo < 65) |>
  rates(mar, ten) |>
  ggplot(aes(mar, ten, fill = rate)) +
  geom_tile(color = "white") +
  scale_fill_viridis_c(labels = scales::label_percent()) +
  labs(title = "Uninsured rate, ages 15-64", x = "Marital status",
       y = "Tenure", fill = "Uninsured", caption = "Figure 9") +
  theme_minimal(12)
```

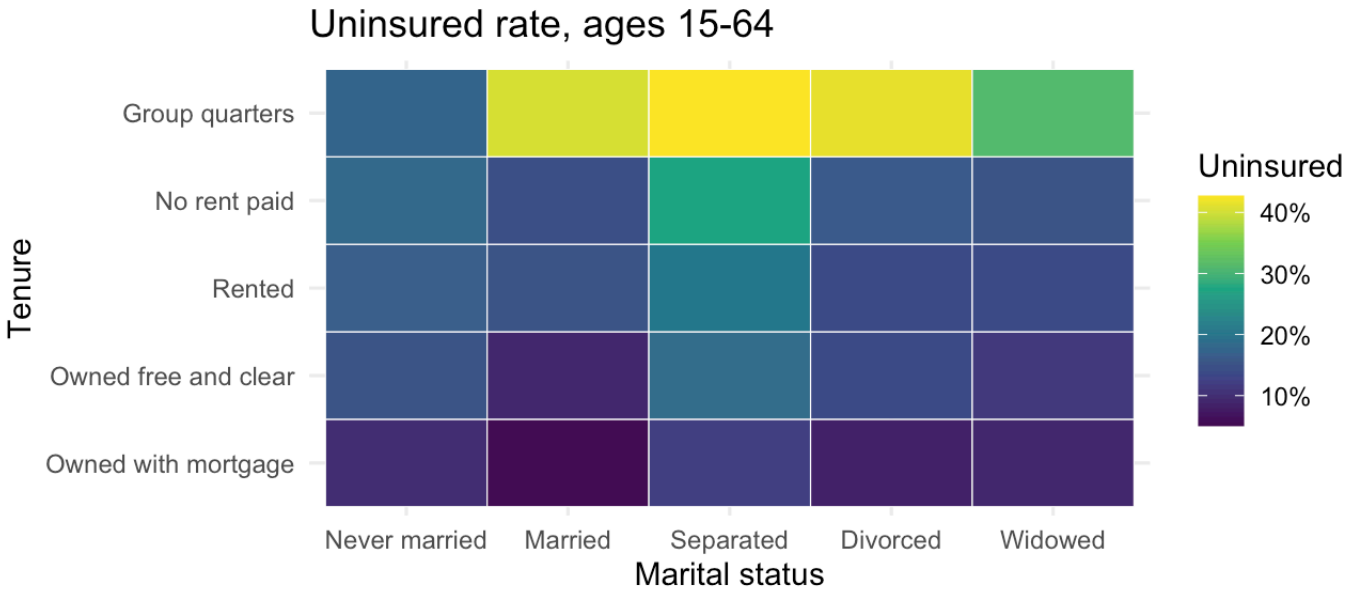


Figure 9

```r
# one point per table cell (marital x coverage x tenure x age)
cells |>
  ggplot(aes(age, count + 1, color = coverage)) +
  geom_point(position = position_jitter(width = 0.25, height = 0,
                                        seed = 1),
             alpha = 0.7) +
  scale_y_log10(labels = scales::label_comma()) +
  scale_color_manual(values = coverage_cols) +
  labs(title = "Estimated people in each table cell", x = "Age",
       y = "People + 1 (log scale)", color = "Coverage",
       caption = "Figure 10") +
  theme_minimal(12)
```



Figure 10